

COMPANION SLIDES

Chapter 4

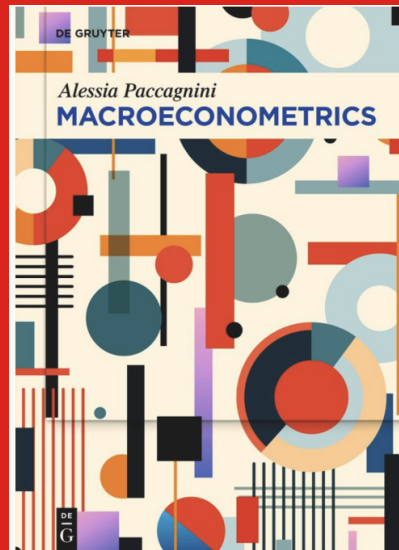
Shock Identification

From Innovations to Structural Shocks

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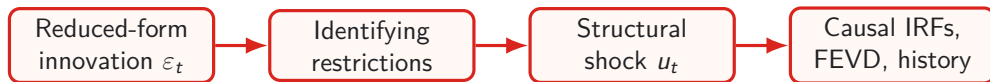
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A reduced-form innovation is observable.

But what economic shock generated it?



- ▶ Formalize the identification problem.
- ▶ Implement recursive / Cholesky identification.
- ▶ Use long-run and short-run restrictions.
- ▶ Understand sign and narrative restrictions.
- ▶ Identify shocks with external instruments.
- ▶ Understand high-frequency identification.
- ▶ Exploit heteroskedasticity and non-Gaussianity.
- ▶ Connect VAR and LP identification.
- ▶ Compute IRFs, FEVD and historical decompositions.
- ▶ Assess robustness across identification schemes.

Reduced-form VAR

$$Y_t = c + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + \varepsilon_t, \quad E(\varepsilon_t \varepsilon_t') = \Sigma_\varepsilon.$$

Structural mapping

$$\varepsilon_t = B_0 u_t, \quad E(u_t u_t') = I_K, \quad \Sigma_\varepsilon = B_0 B_0'.$$

Key takeaway

The reduced form identifies Σ_ε , not a unique structural impact matrix B_0 .

Counting restrictions

B_0 contains K^2 unknowns, while the symmetric covariance matrix supplies only $K(K + 1)/2$ unique equations.

Missing information

We need

$$\frac{K(K - 1)}{2}$$

additional restrictions to achieve point identification.

Any orthogonal rotation Q produces an observationally equivalent impact matrix:
 $\Sigma_\varepsilon = (B_0 Q)(B_0 Q)'$.

Strategy	Information used	Typical question
Cholesky	contemporaneous timing	who can react within the period?
Long-/short-run restrictions	economic theory	which effects are zero now / forever?
Sign restrictions	qualitative theory	what signs should responses have?
External instruments	outside variation	can a proxy isolate the target shock?
Narrative restrictions	historical episodes	what do we know about specific dates?
Statistical identification	volatility / distributions	can data features separate shocks?

Decomposition

$$\Sigma_{\varepsilon} = PP',$$

where P is lower triangular. Setting $B_0 = P$ creates a recursive contemporaneous structure.

- ▶ Simple, transparent and widely used.
- ▶ Ordering encodes economic timing assumptions.
- ▶ Results can be sensitive to ordering: with K variables there are $K!$ possible orderings.

Key takeaway

Cholesky is not “assumption free”: the ordering *is* the identifying assumption.

Core idea

Theory may imply that some structural shocks have no permanent effect on selected variables. These long-run zeros can identify the structural system without restricting short-run dynamics.

- ▶ Classic example: Blanchard–Quah permanent versus transitory shocks.
- ▶ Long-run impact matrix depends on the VAR's cumulative dynamics.
- ▶ Results can be sensitive to lag specification, deterministic terms and weak long-run effects.

Beyond recursive ordering

- ▶ impose selected zero restrictions in the contemporaneous system;
- ▶ allow nonrecursive patterns;
- ▶ motivate restrictions using economic timing or institutional structure.

Applied principle

Write the economic assumptions first, then translate them into restrictions on the structural matrices. Overidentifying restrictions can be tested when the model imposes more restrictions than minimally required.

Replace exact zeros with qualitative implications

$$s_{ijh} [\Theta_h(Q)]_{ij} \geq 0, \quad \tilde{B}_0 = B_0 Q, \quad QQ' = I_K.$$

- ▶ Draw admissible rotations and retain those satisfying the restrictions.
- ▶ Often delivers **set identification**, not a unique structural model.
- ▶ Results depend on which signs, variables and horizons are restricted.
- ▶ Wide identified sets signal limited identifying information.

A proxy for the target structural shock

$$z_t = \alpha u_{1t} + \eta_t.$$

Relevance

$$E(z_t u_{1t}) \neq 0$$

Exogeneity

$$E(z_t u_{jt}) = 0, \quad j \neq 1$$

The chapter also emphasizes lead-lag exogeneity: the instrument should be uncorrelated with past and future structural shocks.

Use narrow event windows

Financial-market surprises around policy announcements can isolate unexpected policy news while reducing contamination from other macroeconomic developments.

- ▶ Commonly used as external instruments for monetary policy shocks.
- ▶ Credibility depends on the event window and interpretation of the surprise.
- ▶ Information effects can complicate the mapping from market surprises to pure policy shocks.

Use historical knowledge

- ▶ constrain the sign of a shock on specific dates;
- ▶ identify episodes in which one shock was dominant;
- ▶ combine narrative restrictions with sign or other restrictions.

Why useful?

Narrative information can shrink a broad identified set by anchoring the model to documented historical episodes.

Heteroskedasticity

If structural shocks change volatility differently across regimes, covariance changes can reveal the mixing matrix.

Non-Gaussianity

Distinctive non-Gaussian shock distributions can separate independent components without conventional zero restrictions.

Key takeaway

These methods shift identifying information from economic restrictions to testable statistical features of the data.

- ▶ Identification determines the structural shock.
- ▶ The dynamic response can then be estimated through a VAR or local projections.
- ▶ External instruments lead naturally to Proxy SVAR or LP-IV.
- ▶ Comparing VAR and LP estimates under the same identification can be a valuable robustness exercise.

Keep the distinction clear

Identification answers “which shock?”; **estimation** answers “what dynamic response?”

Structural IRF

$$\Theta_{h,ij} = e_i' \Phi_h B_0 e_j.$$

- ▶ Φ_h carries the reduced-form propagation mechanism.
- ▶ B_0 maps orthogonal structural shocks into reduced-form innovations.
- ▶ Identification therefore changes the economic interpretation of the same estimated VAR dynamics.

FEVD

How important is shock j for the forecast error variance of variable i at horizon h ?

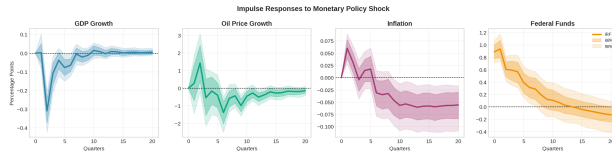
Historical decomposition

How much did each identified structural shock contribute to a particular historical episode?

Identification turns statistical innovations into economically interpretable decompositions.

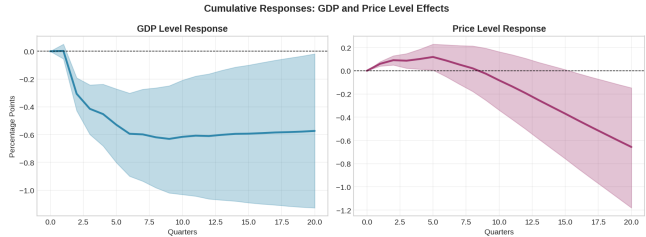
The chapter's comprehensive application identifies US monetary policy shocks using **Cholesky** and **Proxy SVAR** approaches with real data.

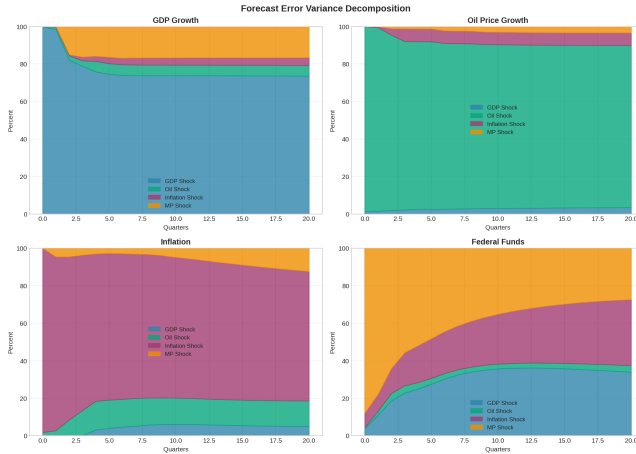
- ▶ compare identification assumptions;
- ▶ trace dynamic responses;
- ▶ quantify uncertainty;
- ▶ assess robustness across methods.



Why cumulate?

For growth rates and flow variables, cumulative IRFs can translate short-run responses into economically meaningful level effects over time.

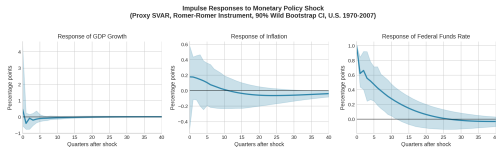




FEVD translates identified shocks into their relative importance for forecast uncertainty.

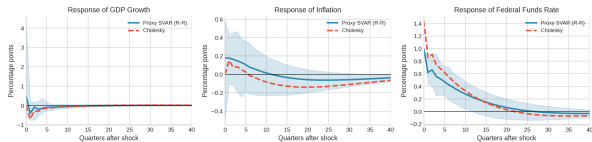


Historical decomposition links structural shocks to particular macroeconomic episodes.



- ▶ Instrument relevance is essential.
- ▶ Weak instruments can produce unreliable structural inference.
- ▶ Bootstrap design must preserve the covariance between residuals and the instrument.
- ▶ Report first-stage diagnostics and uncertainty.

Monetary Policy Shock: Proxy SVAR vs Cholesky
(90% Bootstrap CI, U.S. Data 1970-2007)



Robustness question

Do the economically important features of the response survive when the identifying assumptions change?

Disagreement across methods is informative: it signals that identification assumptions matter for the conclusion.

1. Estimate and diagnose the reduced-form model.
2. State the structural shock of interest.
3. Choose and justify an identification strategy.
4. Check instrument strength or restriction sensitivity.
5. Compute IRFs and uncertainty bands.
6. Add FEVD / historical decomposition where useful.
7. Compare credible alternative identification schemes.

1. Reduced-form VARs do not by themselves deliver causal shocks.
2. Identification requires information beyond the reduced-form covariance matrix.
3. Different strategies encode different economic, historical or statistical assumptions.
4. External instruments are powerful only when relevance and exogeneity are credible.
5. Robustness across defensible identification schemes strengthens structural interpretation.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 4. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add alternative identification examples and orderings.
- ▶ Add your own instruments, event studies or narrative episodes.
- ▶ Add coding demonstrations, derivations and exercises.
- ▶ Ask students to compare conclusions across identification schemes.

Make them yours.

Bayesian VARs

From unrestricted parameter proliferation to disciplined shrinkage

Next question

How can prior information and shrinkage improve estimation when VARs become large and parameter-rich?